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multiple antibiotics and potential transfer of plasmids to other organisms. Proper hand washing is crucial to prevent transmission from patient to patient via medical personnel. Contact isolation should be used for patients colonized or infected with highly antibiotic-resistant *Klebsiella* strains.

*Klebsiella ozaenae* and *Klebsiella rhinoscleromatis* are infrequent isolates that are subspecies of *K. pneumoniae*; however, each is associated with a unique spectrum of disease. *K. ozaenae* is associated with atrophic rhinitis, a condition called ozena, and purulent infections of the nasal mucous membranes. *K. rhinoscleromatis* causes the granulomatous disease rhinoscleroma, an infection of the respiratory mucosa, oropharynx, nose, and paranasal sinuses.

For the otherwise healthy individual, antimicrobial therapy is often unnecessary. *Klebsiella* thrives on a diet high in starch, so a low-starch diet may be helpful. A further caution is that *Klebsiella* thrives on Fructooligosaccharides (FOS) a class of oligosaccharides used as an artificial or alternative sweetener. Antibiotics may be indicated if symptoms are prolonged and in systemic infections. Refer to the bacterial sensitivities to identify the most appropriate pharmaceutical or natural agent.

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#### Proteus species

*Proteus* species are gram-negative bacteria that belong to the Enterobacteriaceae family. *Proteus* species are normal inhabitants of human flora but are listed as dysbiotic flora at 3 - 4+. *P. vulgaris* and *P. mirabilis* are the best-known species of the genus *Proteus*, with *P. mirabilis* being the most common infectious species. *Proteus* is commonly found in contaminated meat, soil, polluted water, sewage and manure, and in the intestinal contents of wild and domestic animals. This bacterium is often found in higher numbers in those who have received antibiotics.

*Proteus* can produce an enterotoxin, and organisms often localize in already damaged tissues in various parts of the body where they may produce an exudative inflammatory reaction. *Proteus* has been associated with diarrheal disease in susceptible individuals and has been implicated as a possible cause of epidemic diarrhea in infants. In addition, *Proteus mirabilis* is a urease-producing organism that contributes to the formation of kidney stones. Urinary tract infections, most common in patients with long-term catheterizations, may lead to kidney failure and may be the source of bacteremia. Infection has been linked to rheumatoid arthritis (antigenic cross-reactivity). There is

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also evidence for osteomyelitis caused by *Proteus* bacteria.

For the otherwise healthy individual, antimicrobial therapy is often unnecessary. Antibiotics may be indicated if symptoms are prolonged and in systemic infections. Refer to the bacterial sensitivities to identify the most appropriate pharmaceutical or natural agent.

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#### Salmonella group

*Salmonella*, a gram-negative rod and member of the Enterobacteriaceae family, is listed as dysbiotic in any amount. The bacteria may be isolated from a wide range of mammals, birds, reptiles, human or animal excrement and contaminated water. The salmonellae are the most diverse of all the Enterobacteriaceae, with more than 2,500 serotypes. *Salmonella* is responsible for foodborne toxic infections due to contaminated raw or undercooked meats, eggs, milk products, and plants used as food. This bacterium may also be found in fish, chocolate, dried pasta, and mayonnaise. Prevention of *Salmonella* infection relies on avoiding contamination by improving hygiene and food handling practices. *Salmonella* infections have more severe effects on infants, the elderly, and immunocompromised individuals.

*Salmonella* can cause either acute gastroenteritis or typhoid fever (typhus). Typhus results from bacterial invasion of the bloodstream. Typically, symptoms of enteritis appear about 12-72 hours following ingestion of contaminated food and last 4-7 days. Symptoms include abdominal pain, diarrhea, vomiting, fever, and possible bloody stools. Most persons recover without treatment. However, in some persons, the diarrhea may be so severe that the patient needs to be hospitalized. In these patients, the infection may cause localized infections (e.g., osteomyelitis or urinary tract infection) or bacteremia. The elderly, infants, and those with impaired immune systems are more likely to have a severe illness. *Salmonella* has also been associated with reactive arthritis. Typhoid fever is a serious bloodstream infection and typically presents with a sustained debilitating high fever and headache, without diarrhea. About 5% of patients clinically cured of *S. typhi* remain carriers for up to one year.

Treatment of enteritis with antibiotics is not advised because it is ineffective, increases the chance of a carrier state developing, and may promulgate the further spread of antibiotic resistance.

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Antibiotics may be indicated if symptoms are prolonged and in systemic infections. Refer to the bacterial sensitivities to identify the most appropriate pharmaceutical or natural agent.

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#### Cultured Yeast

Yeast, such as *Candida* are normally present in the GI tract in very small amounts. Many species of yeast exist and are commensal; however, they are always poised to create opportunistic infections and have detrimental effects throughout the body. Factors that contribute to a proliferation of yeast include frequent use of wide-spread antibiotics/low levels of beneficial flora, oral contraceptives, pregnancy, cortisone and other immunosuppressant drugs, weak immune system/low levels of sIgA, high-sugar diet, and high stress levels.

When investigating the presence of yeast, disparity may exist between culturing and microscopic examination. Yeast grows in colonies and is typically not uniformly dispersed throughout the stool. This may lead to undetectable or low levels of yeast identified by microscopy, despite a cultured amount of yeast. Conversely, microscopic examination may reveal a significant amount of yeast present, but no yeast cultured. Yeast does not always survive transit through the intestines rendering it unviable for culturing. Therefore, both microscopic examination and culture are helpful in determining if abnormally high levels of yeast are present.

#### Microscopic yeast

Microscopic examination has revealed yeast in this stool sample. The microscopic finding of yeast in the stool is helpful in identifying whether the proliferation of fungi, such as *Candida albicans*, is present. Yeast is normally found in very small amounts in a healthy intestinal tract. While small quantities of yeast (reported as none or rare) may be normal, yeast observed in higher amounts (few, moderate to many) is considered abnormal.

An overgrowth of intestinal yeast is prohibited by beneficial flora, intestinal immune defense (secretory IgA), and intestinal pH. Beneficial bacteria, such as *Lactobacillus* colonize in the intestines and create an environment unsuitable for yeast by producing acids, such as lactic acid, which lowers intestinal pH. Also, *Lactobacillus* is capable of releasing antagonistic substances such as hydrogen peroxide, lactocidin, lactobacillin, and acidolin.

Many factors can lead to an overgrowth of yeast including frequent use of antibiotics (leading to insufficient beneficial bacteria), synthetic corticosteroids, oral contraceptives,

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and diets high in sugar. Although there is a wide range of symptoms which can result from intestinal yeast overgrowth, some of the most common include brain fog, fatigue, recurring vaginal or bladder infections, sensitivity to smells (perfumes, chemicals, environment), mood swings/depression, sugar and carbohydrate cravings, gas/bloating, and constipation or loose stools.

A positive yeast culture (mycology) and sensitivity to prescriptive and natural agents is helpful in determining which anti-fungal agents to use as part of a therapeutic treatment plan for chronic colonic yeast. However, yeast are colonizers and do not appear to be dispersed uniformly throughout the stool. Yeast may therefore be observed microscopically, but not grow out on culture even when collected from the same bowel movement.

#### Fecal Lactoferrin

The level of fecal lactoferrin, a biomarker of serious gastrointestinal inflammation, is abnormally high in this fecal sample. Fecal lactoferrin is elevated in association with Inflammatory Bowel Disease (IBD) such as Ulcerative Colitis (UC) or Crohn's Disease (CD)[1,2], but NOT Irritable Bowel Syndrome (IBS)[1,3]. Therefore, assessment of fecal lactoferrin levels enables distinction between IBD and non-inflammatory IBS. Such distinction is critical because, although both IBD and IBS may share some common symptoms such as diarrhea, abdominal cramping and weight loss, the diseases are treated quite differently. IBD may become life threatening, requires life long treatment and possibly surgery. In contrast, IBS is often effectively treated with dietary restrictions, stress reduction and medication.

Gastrointestinal inflammation associated with IBD is associated with increased infiltration of activated neutrophils into the mucosa and increased release of lactoferrin into the gut[1,4,5]. Patients with inflammation of the GI tract, such as IBD (but not IBS), exhibit elevated lactoferrin concentrations in the feces[1].

Clinical studies have shown that fecal lactoferrin levels of healthy persons are similar to IBS patients, but markedly increased in patients with active IBD[1,3]. Patients with IBD oscillate between active and inactive disease states, and fecal lactoferrin levels increase 2-3 weeks prior to onset of clinical symptoms[6]. During remission and effective treatment, fecal lactoferrin decreases significantly. Therefore disease activity, and efficacy of treatment can be monitored by following fecal lactoferrin levels. The test can be ordered separately to track disease activity in patients with IBD.

Moderately elevated levels of fecal lactoferrin can occur, with fecal red blood cells and leukocytes, in association with invasive enteropathogens [7,8]. Such levels would be expected to be much lower than those associated with the active phase of IBD. Therefore, with moderately elevated levels of fecal lactoferrin, one should check for the presence of enteropathogens (eg. Shigella, Campylobacter, Clostridium difficile).

Guidelines for interpreting the results of this test are provided by the results of a large multi-center clinical study which evaluated fecal lactoferrin levels in 180 patients suffering with IBS and IBD (UC and CD) compared to 56 healthy controls.

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| GROUP            | # of SPECIMENS | FECAL LACTOFERRIN  |
|------------------|----------------|--------------------|
|                  |                | mean mcg/ml +/- SE |
| Inactive UC      | 41             | 67 +/- 24          |
| Active UC        | 31             | 815 +/- 789        |
| Inactive CD      | 26             | 239 +/- 83         |
| Active CD        | 51             | 672 +/- 242        |
| IBS              | 31             | 1.3 +/- 0.3        |
| Healthy Controls | 55             | 1.6 +/- 0.4        |

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#### Secretory IgA (sIgA)

The concentration of sIgA is abnormally high in this fecal specimen. Immunological activity in the gastrointestinal tract can be assessed using secretory immunoglobulin A (sIgA). Secretory IgA is the predominant antibody or immune protein the body manufactures and releases in external secretions such as saliva, tears, and milk [1]. It is also transported through the epithelial cells that line the intestines out into the lumen. Secretory IgA represents the first line of defense of the GI mucosa and is central to the normal function of the GI tract as an immune barrier [1]. As the principal immunoglobulin isotype present

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in mucosal secretions, sIgA plays an important role in controlling intestinal milieu which is constantly presented with potentially harmful antigens such as pathogenic bacteria, parasites, yeast, viruses, abnormal cell antigens, and allergenic proteins [1]. Secretory IgA antibodies exert their function by binding to antigenic epitopes on the invading microorganism limiting their mobility and adhesion to the epithelium of the mucus membrane [2]. This prevents the antigens from reaching systemic circulation allowing them to be excreted directly in the feces.

Elevated fecal sIgA is an appropriate response to an antigenic presence. Microbial and microscopic studies of the stool are useful in identifying if bacteria, yeast, or parasites are present. Eradication of the pathogenic microorganisms will bring sIgA back down into the normal range. Elevated sIgA levels have been observed in the absence of bacteria, yeast or parasites, in individuals with atopic conditions such as food allergies, urticaria, and dermatitis.

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#### Short Chain Fatty Acids (SCFAs)

The relative and/or total amounts of SCFAs are abnormal in this specimen. In infants, microbial colonization and subsequent SCFA production is a gradual process which is largely determined by environmental exposure, maternal gut microflora, breastfeeding, and possibly genetics [1]. The establishment of an adequate distribution of healthy flora in the gut is crucial in the health of both infants and adults because of the "competitive exclusion" process of dysbiotic flora [1]. Healthy microflora, such as *Lactobacillus* and *Bifidus* generate large amounts of SCFAs (acetic,

propionic, butyric, and valeric) which decrease the pH of the intestine and therefore make the environment unsuitable for pathogens, including bacteria and yeast [1]. SCFAs are the end product of the bacterial fermentation process of dietary fiber by beneficial flora in the gut and play an important role in the prevention of intestinal dysbiosis [1].

The amount and types of SCFAs produced by colonic bacteria will depend on factors such as type of fiber consumed and overall intestinal health. For example, more SCFAs in general are produced by gums and pectins than oat fiber or corn bran; more propionate and butyrate are produced by bacterial activity on gums than pectins [2]. Antibiotic-induced diarrhea may be secondary to decreased colonic fermentation of carbohydrates and decreased overall production of SCFAs [3].

Studies show that SCFAs have numerous implications in gut physiology and health. SCFAs decrease inflammation, stimulate healing, and contribute to normal cell metabolism and differentiation [4]. Acetate, propionate and butyrate have been demonstrated to potentially improve the microcirculation in the intestinal mucosa thereby improving its growth and repair [5,6]. Rectal irrigation with SCFAs can result in improvement of ulcerative colitis [7]. Butyrate and propionate contribute to apoptosis of colorectal cells by increasing the production of reactive oxygen species in the gut [8]. Butyrate also has a positive effect on the differentiation of colonocytes, and this may account for the protective effect of dietary fiber against colorectal cancer [9]. Probiotics and increased dietary fiber can improve/normalize SCFA status.

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#### Fecal Calprotectin

The level of fecal Calprotectin is higher than expected. Elevated fecal Calprotectin and Lactoferrin levels indicate the presence of neutrophils and inflammation in the gastrointestinal (GI) mucosa. Calprotectin and Lactoferrin facilitate differentiation between irritable bowel syndrome (IBS) and

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inflammatory bowel disease (IBD). IBD includes autoimmune conditions such as Crohns disease and ulcerative colitis (UC); these conditions may become life-threatening and require lifelong treatment.

For patients 4 years old to adults (Manz 2012; Fagerberg 2005):

High levels of Calprotectin ( $> 200 \mu\text{g/gm}$ ) are associated with active IBD and gastrointestinal inflammation; elevation may also occur due to bacterial infection, colitis or sometimes, cancer. Fecal Calprotectin should be reassessed after about 4 weeks for confirmation.

Moderate calprotectin ( $50\text{-}200 \mu\text{g/gm}$ ) is an indicator of chronic inflammation. Inflammation at this level may be due to IBD in remission or inflammation caused by non-steroidal anti-inflammatories (NSAIDs). Levels should be reassessed after about 4 weeks. Low levels of Calprotectin ( $< 50 \mu\text{g/gm}$ ) are usually associated with viral GI infections or non-inflammatory bowel conditions such as IBS.

Multiple studies have shown fecal Calprotectin and Lactoferrin to be equivalent with respect to clinical sensitivity and specificity. Studies suggest that Calprotectin may correlate more closely with histological (cell microscopy) findings. Lactoferrin may correlate better to macroscopic (endoscopy) findings, and may be the better indicator of impending relapse, elevating 2-3 weeks prior to clinical symptoms.

Chronic inflammation of the gastrointestinal mucosa contributes to symptoms of IBD. Chronic stress is known to contribute to symptom flare-ups and increased inflammation. Liver disease or the use of aspirin or nonsteroidal anti-inflammatory (NSAID) medications may elevate Calprotectin levels. Fecal Calprotectin levels may also be increased in newborns.

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